

Inventory Performance & Replenishment Brief

Park Royal Depot (London) — 12-Month Inventory Optimization Analysis

1. Executive Summary

PHARx faced severe capital constraints at its Park Royal Depot due to two competing issues: high-cost specialty lines expiring on shelves, and common daily medications suffering frequent stockouts. Using Google BigQuery, 12 months of daily sales and inventory telemetry were analyzed to build a demand driven replenishment model. The model safely prunes excess holding values without introducing safety deficits, keeping daily fulfillment fully business-as-normal.

£372,636.13 Capital Released	6.0% Efficiency Gain	0.0 Fulfillment Risk
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2. Financial Impact Breakdown

The model concentrates capital recovery entirely in high-cost, slow-moving specialty drug lines (Oncology and Immunology), which represent 99.3% of total savings, while keeping everyday maintenance categories secure.

Category	Current Value	Modelled Value	Net Savings
Oncology Lines	£3.19M	£3.00M	£190,000.00
Immunology Lines	£2.87M	£2.69M	£180,000.00
Other Categories*	£0.19M	£0.17M	£2,636.13

**Note: Other Categories include Cardiology, Diabetology, and General Antibiotics aggregated to match slide definitions.*

3. Model Core Mechanics

Before placing any order, the system extracts and considers three variables: product consumption speed (avg_units_sold_per_day), expiration proximity, and supplier delivery turnaround. From these, it automates two core metrics:

- **Dynamic Ordering Cycle (place_order_every_x_days):** Calculates custom timeline windows to prompt shipments right before warehouse safety buffers deplete.
- **Responsive Restock Volume (order_quantity):** Defines the exact volume ceiling required to align replenishment with real-time demand, removing asset accumulation.

4. Strategic Action Plan

01 | Continuous Model Execution

Embed query procedures directly into standard workflows to automatically adjust procurement as daily sales speeds and supplier lead times shift over time.

02 | Strategic Capital Reinvestment

Reallocate the unlocked £372,636.13 directly into system automation and inventory expansion.

03 | Demand-Driven Automation

Scale the model into an auto-ordering tool that places requests driven exclusively by live product demand, while strictly calculating individual product expiry dates, lead times, and safety buffers.

Automated Expansion Signaling: When the auto-ordering tool flags that optimal order volumes cannot be received due to space constraints, it serves as an objective, data-backed signal that physical facility expansion is viable—directly funded by the capital saved from the initial optimization.